

ASSESSMENT TOOL – 2

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO4	Assessment	Industry Problem Solving
Weightage		Total Marks	

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Section / Batch	ECE-2024	Date of Submission	14/08/2026
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

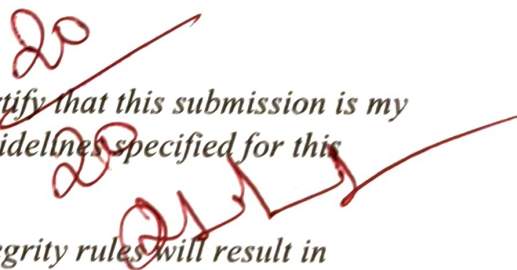
Total Marks		Obtained Marks	
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I Cindy R / 192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

o Person Startup

Answer: star Topology

A star topology is ideal because all devices are connected to a central managed switch. Cat6 UTP cables provide Gigabit Ethernet, while a router/firewall provides Internet access, DHCP, NAT, and security. A Wi-Fi access point supports wireless devices. If one cable or workstation fails, the remaining devices continue working. It is easy to manage, troubleshoot and expand.

Key Points: Low cost, easy maintenance, fault isolation, scalable.

Bank Requiring Near-Zero Downtime

Answer: Partial Mesh / Full Mesh Technology

A partial mesh topology is suitable for a bank because it provides multiple paths between critical network devices. Redundant routers, switches, firewalls and links ensure that failure of one component does not stop network operation. Fiber-optic links provide high bandwidth and low latency. Routing protocols such as OSPF or BGP can automatically select alternative paths.

Key Points: High reliability, redundancy, fault tolerance, fast recovery.

3. Small classroom Lab with Tight Budget.

Answer: Bus Topology.

A bus topology is suitable when cost is the main concern. All computers share a common backbone cable and require less networking hardware. In a traditional implementation, coaxial cable with BNC connectors and 50-ohm terminators is used. However, modern Ethernet networks generally use switches rather than true bus topology.

Key Points: Low cost, simple design, but cable failure can affect the entire network.

Large University Campus

Answer: Hybrid Topology.

A hybrid topology is best because a university has many buildings and different networking requirements. A fiber-optic ring or partial mesh can be used for the backbone, while star topology can be used inside individual buildings. Core, distribution and access layers provide scalability and easier management. Fiber links connect buildings while Cat6 cables connect end devices.

Key Points: Scalable, reliable, flexible, suitable for multiple buildings.

Company with Multiple Departments

Answer: Hierarchical Topology with VLAN's

A hierarchical network can be designed using access, distribution, and core layers. Each department is assigned a separate VLAN, such as HR, Finance, sales and IT. Layer 2 switches connect departmental devices, while a Layer 3 switch or router performs inter-VLAN routing. A firewall provides security and Internet access.

Key Points: Hierarchical topology, separate VLANs, firewall, subnetting.

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